

Physics-guided machine learning predicts the planet-scale performance of solar farms with sparse, heterogeneous, public data

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HIGHLIGHTS

- Introduces a data-efficient, physics-guided machine learning approach for solar farms.
- The model predicts global annual yield potential with high accuracy, achieving an error of less than 6 % compared to a detailed physics-based model.
- Monthly yield data from as few as five representative locations can predict global yield, significantly reducing data requirements for machine learning training.
- Introduces PVZones, a PV-specific climate map, to facilitate the creation of representative datasets for data-efficient training.
- Demonstrates how real-world PV performance data can be effectively utilized for yield prediction despite its geographic sparsity and heterogeneity.

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ABSTRACT

The photovoltaics (PV) technology landscape is evolving rapidly. To predict the potential and scalability of emerging PV technologies, a global understanding of these systems' performance is essential. Traditionally, experimental and computational studies at large national research facilities have focused on PV performance in specific regional climates. However, synthesizing these regional studies to understand the worldwide performance potential has proven difficult. Given the expense of obtaining experimental data, the challenge of coordinating experiments at national labs across a politically divided world, and the data privacy concerns of large commercial operators, a fundamentally different, data-efficient approach is desired. Here, we introduce a physics-guided machine learning (PGML) approach for PV to demonstrate that: (a) the world can be divided into a few PV-specific climate zones, called PVZones, illustrating that the relevant meteorological conditions are shared across continents; (b) by exploiting the climatic similarities, high-quality monthly energy yield data from as few as five locations can accurately predict (with a root mean square error of less than 8 kWh m⁻²) global yearly energy yield potential at high spatial resolution. Moreover, by homogenizing noisy, heterogeneous public PV performance data, the global energy yield can be predicted with less than 6 % relative error compared to physics-based simulations, provided that the dataset is representative. This novel data-efficient PGML scheme for PV is independent of both PV technology and farm topology, allowing it to adapt seamlessly to emerging PV technologies and farm configurations. The results pave the way for physics-guided, data-driven collaboration between national policymakers and research organizations in developing efficient decision support systems to accelerate PV deployment worldwide.

1. Introduction

The fast-paced advancement of material platforms such as silicon, perovskite, and cadmium telluride (CdTe), along with developments in cell and module technologies, such as passivated emitter and rear contact (PERC), bifacial, tandem, etc., alongside various farm layouts such as fixed-tilt, tracking, vertical, floating, and agrivoltaics, has generated a pressing demand to assess the energy yield (EY) and economic feasibility

of different technological choices and farm arrangements across diverse global regions. After all, modern solar farms require significant capital expenditure, and incorrect decisions have long-term financial, political, and public perception implications.

Several authors have previously reported the global yield performance potential of various solar cells, from single-junction silicon [1] to multi-terminal perovskite-silicon tandem cells [2,3], leading to system-scale techno-economic analyses [4,5]. Such analyses rely

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List of symbols

M	Monthly yield
Y	Yearly yield
I_{GHI}	Global horizontal irradiance
T_{amb}	Ambient temperature
k_t	Clearness index
S_i	Simulated PV dataset
F	Homogenized outdoor PV dataset
F_s	Sampled homogenized outdoor PV dataset

F_s^*	Sampled outdoor PV dataset with synthetic data fusion
X	PGML input variable
E	Percentage error
z	PVZones zone
N	Number of observations in a dataset
M_{field}	Measured monthly yield of outdoor PV system
M_{field}^*	Scaled measured monthly yield of outdoor PV system
M_{sim}	Simulated monthly yield of outdoor PV system
M_{PGML}	Monthly yield predicted by the PGML model
s	Scaling factor

on a number of commercial or free physics-based models, such as PVSyst, Bifacial Radiance, PlantPredict, and PVMAPS. While these models are widely used and essential for planning and decision-making, they become computationally expensive when applied to regional or global-scale analyses. Currently, predicting the energy yield of solar farms relies on either time-intensive computational modeling using proprietary software operated by experts from consulting firms or on aggregating scattered field reports from highly instrumented test centers in a few technologically advanced countries. An alternative framework leveraging machine learning (ML) for fast model prediction and rapid experimental validation could revolutionize solar development and deployment by enabling independent PV performance projections and validation worldwide. However, due to the high cost of obtaining experimental data, the difficulty of coordinating experiments across national labs in a politically divided world, and data privacy concerns of large commercial operators, a more data-efficient and physically consistent machine-learning approach is needed.

In recent years, machine learning models have demonstrated unprecedented success, especially large language models, in representation learning problems. Training highly parameterized models with large amounts of data is computationally expensive and unsustainable over the long term due to increasing carbon footprint [6]. Furthermore, the models may still produce results that are inconsistent with physical laws [7,8], which may prevent their direct use in decision support systems. By leveraging the physical understanding of the data and processes being modeled, a physics-guided machine-learning (PGML) approach reduces the dependency on large datasets and improves prediction accuracy [7]. As such, PGML could transform the decision-making process by significantly accelerating the analysis and design time by orders of magnitude. Recognizing this potential, PGML approaches are being explored across all system scales, from cells to farms, including material discovery [9], solar cell optimization [10,11], solar resource prediction [12], module degradation mode identification [13], and solar farm performance prediction [10,14]. Over the years, a substantial amount of PV performance data has been published in the literature and is accessible through public data dashboards from PV systems deployed at airports, universities, and weather stations. However, because the combined data encompasses a mix of materials, module types, tilt angles, farm configurations, climate conditions, and time granularity, addressing the heterogeneity in these datasets to train a reliable ML model remains a challenge. This paper presents a data-efficient PGML approach for predicting global solar farm yield potential, even when the data is heterogeneous.

Previous work by Patel et al. [14] proposed the possibility of a data-driven approach to PV yield prediction using high-quality experimental data. They demonstrated that a physics-guided ML-based functional approximation could reduce global-scale prediction time by orders of magnitude, from hours to seconds. Remarkably, the global energy yield potential was predicted with less than 10 % error using a relatively small dataset of simulated yearly yield observations from 100 randomly selected locations worldwide. Despite the impressive results, a

manufacturer may find it impractical or costly to install newly developed modules to collect year-long yield data from hundreds of locations worldwide. In the context of PV, data-efficient learning is especially important due to the limited availability of publicly accessible, high-quality, long-term global-scale PV performance data that spans a wide range of climatic conditions worldwide.

In this work, we show how the physical understanding of PV systems can inform the design of data-efficient, generalizable PGML models for global PV performance prediction. Specifically, it will be demonstrated that the EY potential of solar farms across the world can be predicted with high accuracy (≤ 6 % error) at a high spatial resolution ($0.5^\circ \times 0.5^\circ$), even when using a model trained on geographically sparse, heterogeneous, and public datasets. The data quantity and quality requirements for training a global-scale model are systematically addressed. In particular, it is demonstrated that diverse sampling of data for physically relevant variables is key to training such models. To guide such sampling, PVZones, a PV-specific climate map, is introduced, facilitating the construction of small yet representative datasets for training a global-scale model. The strength of our approach lies in recognizing climatic similarities worldwide: PV data collected from one site can represent conditions in other locations, enabling performance predictions for many geographically distant areas. Moreover, the proposed methodology is problem-agnostic, meaning any technology metric that depends on local climate (e.g., reliability of automotive electronics) can be analyzed by dividing the world into technology-specific climate zones.

The four central ideas introduced in the paper are as follows:

- *Spatial diversity sampling.* If the world can be segmented into regions defined by effectively similar input variables, then relatively few (functionally important) samples from each of these regions would train the ML model far more efficiently than random sampling possibly could.
- *Temporal equivalence sampling.* The values of those variables seen at a location depend on the position of the sun, and thus, change monthly throughout the year. For example, West Lafayette feels like Arizona during the summer, and Alaska during the winter! Therefore, an ML model trained with monthly yield (M) data from a given location could predict the yields at many other locations during different months.
- *Utilizing heterogeneous public datasets.* The predictions of a PGML trained on large-scale synthetic data can be used to efficiently homogenize publicly available heterogeneous PV performance datasets. These datasets can then be used to train the PGML model based on experimentally obtained field data.
- *Physics-guided data fusion.* A large quantity of field data alone is insufficient to train an ML model that can be generalized globally if there are gaps in representativeness. Combining field data with gap-filling synthetic data creates a more representative dataset for training machine learning models to predict global PV performance.

The paper is organized as follows. Section 2 introduces our PGML approach to global solar farm yield prediction and the key hypothesis. Sections 3 and 4 demonstrate the performance of the model when trained with simulated and field-obtained data, respectively. Finally, Section 5 summarizes the results and discusses their broader implications.

2. Physics-guided ML models for solar farms

Multiple approaches exist to develop an ML model guided by physics [15], with the general goal being to train a model that makes physically consistent predictions. One approach is to train the model with data obtained from numerical and/or physical experiments. This training data, consisting of physically relevant inputs and corresponding outputs, embodies the physics of the data-generating process. In this approach, a clear understanding of the physics behind the modeled process is essential, as it directly informs the choice of variables. When the input domain of the problem is sufficiently represented in the training dataset, the ML model (e.g., Neural Network, Random Forest, etc.) can predict solutions for unknown inputs. This is the approach we adopt in this work.

2.1. Choosing physically-relevant variables

In this work, we seek to train a PGML model that predicts the global annual energy yield potential (Y) of solar farms (shown in Fig. 2). Y is the maximal DC yield potential of a solar farm employing a given technology, neglecting any time-dependent losses or limiting factors (e.g., module degradation, soiling, inverter losses). For a given farm design, the yield potential can be computed with complex physics-based models that simulate optical, thermal, and electrical conditions in the solar farm [16].

The experience of developing such forward models enables us to refine the input space and identify variables that are physically meaningful for a specific problem, i.e., determining Y . As mentioned, a small set of meteorological variables determines the Y potential at a given location: the global horizontal irradiance (I_{GHI}), the ambient temperature (T_{amb}), and the clearness index (k_t) measuring the cloud conditions (i.e., direct vs. diffused sunlight contributions). Therefore, the input variables for PGML are $X = (I_{\text{GHI}}, T_{\text{amb}}, k_t)$. It is easy to show that other features are either correlated (e.g., module temperature, T_{mod}) or redundant (e.g., physical coordinates). It is important to note that the variables chosen (X) should be physically significant for the target variable of interest. For instance, if the lifetime degradation of solar modules owing to solder bond failure were the target variable, then the range of temperature fluctuations (ΔT) would be a physically relevant variable to predict the phenomenon [17].

Having identified the physically significant variables (i.e., $I_{\text{GHI}}, T_{\text{amb}}, k_t$), the problem of determining Y can be cast as a regression problem. Let the monthly yield potential (M) be expressed as a function

$$M = f(I_{\text{GHI}}, T_{\text{amb}}, k_t) \quad (1)$$

The desired quantity, Y potential, is then readily calculated for each location by summing the predicted M , i.e., $Y = \sum_{i=1}^{12} M_i$.

2.2. PGML model

The machine learning model for yield prediction was a feed-forward neural network with two hidden layers (10 hidden neurons on each layer). Hereon, we will refer to this neural network as the PGML model. The network was implemented in MATLAB utilizing the Deep Learning Toolbox. During training, Bayesian regularization (trainbr) was used to obtain the optimal number of parameters while penalizing overfitting. A training dataset (D_i) consists of N monthly average values of $I_{\text{GHI}}, T_{\text{amb}}, k_t$, and the corresponding M at different locations across the globe. M was obtained by simulations or by homogenizing the measured monthly yields of public PV systems.

For a model to achieve global generalizability, its training dataset D_i must encompass diverse combinations of input variables ($I_{\text{GHI}}, T_{\text{amb}}, k_t$) as observed across various regions worldwide. Fortunately, these meteorological variables are often measured with high spatiotemporal resolution, thanks to the ever-expanding databases of satellite observations [18]. In this study, we used NASA's POWER [19] climatological dataset (see Appendix).

However, it is unclear what quantity of observations (N) will be necessary to accurately approximate the function in Eq. (1) globally. Collecting a substantial amount of training data becomes impractical and cost-prohibitive because M values must be either simulated or measured from fielded PV systems. To tackle this issue, we will leverage the climatic similarity between different regions worldwide to create small yet representative datasets.

2.3. PV-specific climate zones enable diversity sampling

Since ancient times, people have understood that Earth's climate can be categorized based on shared climatic characteristics [20]. Several classification systems have been devised, e.g., Köppen–Geiger (K–G) [21], Thornthwaite [22], etc., each using different sets of variables to explain climatic similarity and variability across the planet. The set of climatic variables is chosen based on their relevance to the issue at hand. For example, the popular K–G system uses precipitation and temperature to explain patterns in vegetation growth. Due to its familiarity, K–G system has been extended to PV by incorporating irradiance [23]. However, it considers variables that may only influence the yield indirectly: cloud conditions in general play a more direct role in determining PV performance potential compared to precipitation.

Instead, by explicitly considering variables relevant to the PV yield problem, we can create climate maps tailored to the problem. Particularly, we can identify the various regions in the world sharing similarities in climatic features that specifically influence PV energy yield. Naturally, this also identifies the unique climates that need to be represented in a dataset if we hope to train the model for global-scale predictions. Here we introduce PVZones (PVZ), a climate map of PV energy data, that divides the world into seven PV energy yield specific zones (z) sharing similarity in $I_{\text{GHI}}, T_{\text{amb}}$, and k_t . Fig. 1(b) illustrates the seven geographical zones (z). The map was constructed by the k-means ($k = 7$) algorithm to cluster the annual average values of the three key features identified. We observe that larger countries (e.g., the United States and China) possess a greater diversity of zones, as expected. While the influence of local climate is well understood, PVZones provide a systematic and targeted approach to planning PV testbeds.

2.4. Hypothesis: sparse data can train a global scale ML model

In developing the method, we hypothesized that by utilizing this climatic equivalence, the global annual yield potential can be estimated using data from a limited number of strategically selected test sites. To test the hypothesis, we evaluated two categories of datasets, representing the two extremes of controlled experiments and data quality. The first category comprises synthetic datasets where M are estimated by PVMAPS [24], a physics-based solar farm model, representing strictly controlled numerical experiments (shown in the top branch in Fig. 1(a)). The second category consists of publicly sourced datasets of outdoor PV performance with minimal control and little to no oversight. The inherent and potentially unexplained heterogeneity of these datasets makes them difficult to use directly for training. This led us to develop a physics-based approach to homogenize the dataset before it can be used for model training (shown in Fig. 1(a) bottom branch). Realistic experimental data will have some level of control between these two extremes.

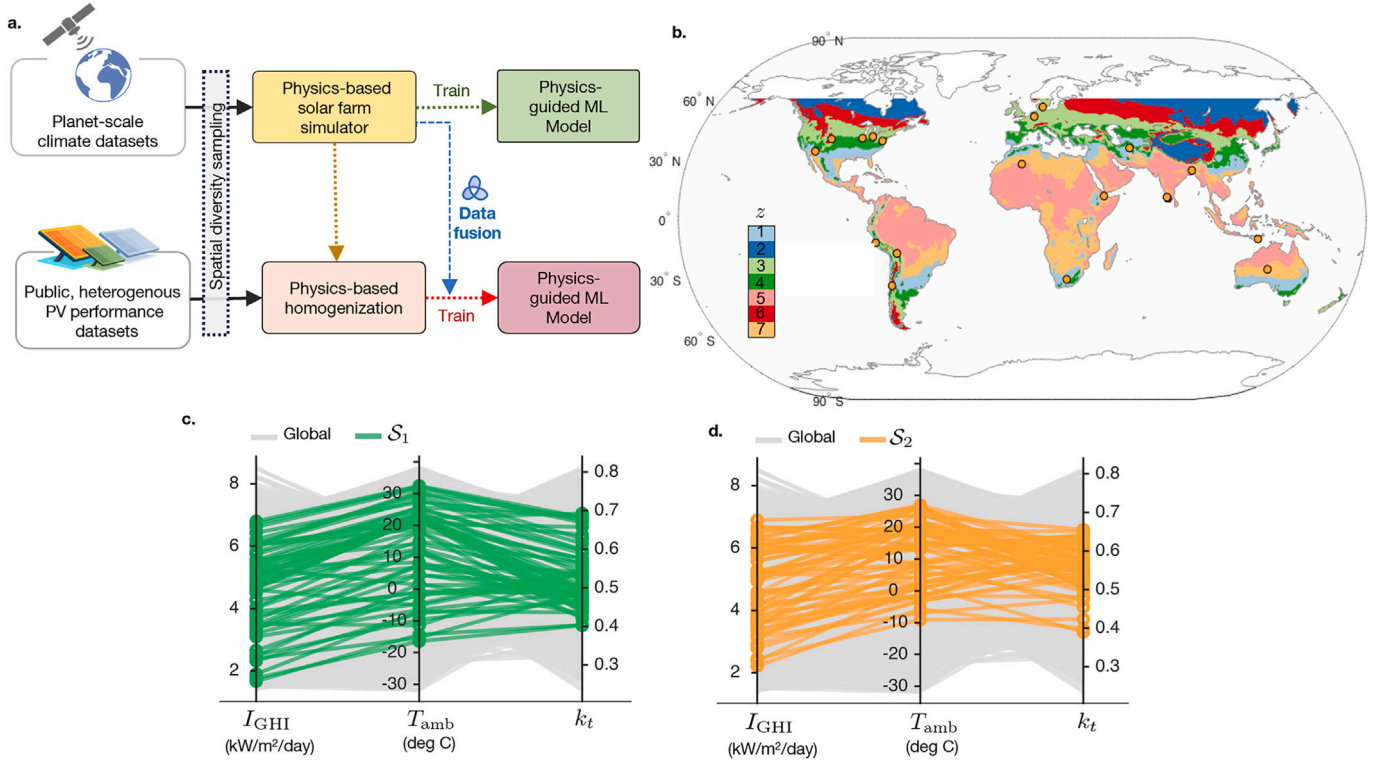


Fig. 1. (a) Overview of the PGML methodology for solar farms leveraging spatial diversity sampling. (b) PVZones (PVZ): geographical distribution of physically significant variables (I_{GHI} , T_{amb} , k_t) zoned with k-means clustering algorithm ($k = 7$). The map illustrates the similarity between the features determining yield potential across continents, which can be leveraged to construct a training dataset with an optimal set of test locations. (c and d) The gaps in the representativeness of a dataset can be visualized by overlaying its observations (green and orange strings) on the global data (gray strings) in a parallel coordinates plot, which displays the range of monthly average values for the variables observed worldwide.

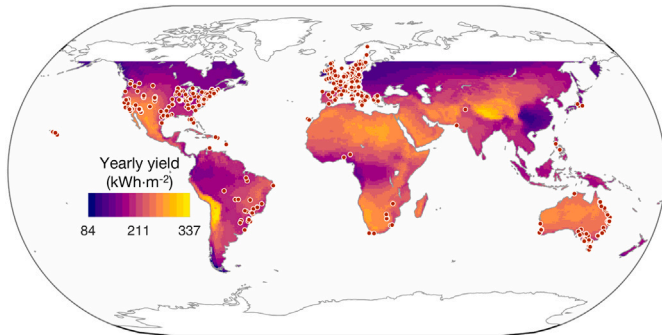


Fig. 2. Worldwide yearly yield potential of fixed-tilt, monofacial solar farms determined by worldwide gridded simulation in PVMAPS, a physics-based solar farm simulator. These results serve as the benchmark to test the data-efficient PGML developed in this paper through spatial diversity sampling. The red dots indicate the locations of over 3000 PV systems with varying capacities, primarily concentrated in North America and Europe. Despite the large quantity of publicly available data, the dataset lacks representation for certain PV zones (cf. Fig. 1(b)).

3. Synthetic PV datasets: strictly controlled numerical experiments

First, we tested whether a PGML model can predict the global yield potential when trained on simulated PV yield from limited locations. For this purpose, two synthetic training datasets were created, each containing M from only five locations. Dataset S_1 contained data from five different sites in five different zones in PVZ. For dataset S_2 , the five selected sites were the locations of the US Department of Energy's (DOE)

regional PV test centers located in the continental US. For each location, M of monofacial solar farms were obtained by the physics-based solar farm simulator PVMAPS. Fig. 3(a–b) show the simulated M at each test site. The simulated farms employed identical modules and had optimal fixed-tilt designs (see Appendix). Both datasets contained only 60 observations of monthly average I_{GHI} , T_{amb} , k_t , and corresponding M .

3.1. Predicting the global performance from five locations

After training the PGML model with S_1 and S_2 , Y was computed for the globe. The model's ability to generalize was evaluated by comparing the predicted Y with the results obtained from a global physics-based simulation. Fig. 3(c–d) and (e–f) show the prediction accuracies of the models trained on S_1 and S_2 , respectively. For the model trained on S_1 , we find that predictions for 95 % locations are within 9 kWh m⁻² of the simulated Y , with higher errors typically occurring in zones that were not represented in the training dataset (e.g., Northern China, Mongolia). Fig. 3(d) shows that the model has good accuracy: the overall root mean squared error (RMSE) was only 3.85 kWh m⁻². Fig. 3(e–f) shows the prediction accuracy for the model trained on S_2 dataset. Fig. 3(e) shows that, despite the sites being confined to the continental US, the model is still able to predict worldwide Y potential for 95 % of locations with less than 16 kWh m⁻² absolute error and a RMSE of 7.62 kWh m⁻². Again, the errors are higher primarily at locations not represented in the dataset. To see why S_1 outperforms S_2 , we look at the parallel coordinates plots in Fig. 1(c–d) where each connected string represents an observation. It can be seen that S_1 , consisting of data from sites chosen randomly from different climate zones, has a better representation of the input domain compared to S_2 .

These results are quite remarkable because they show that a diverse and high-quality experimental dataset from a small number of test sites

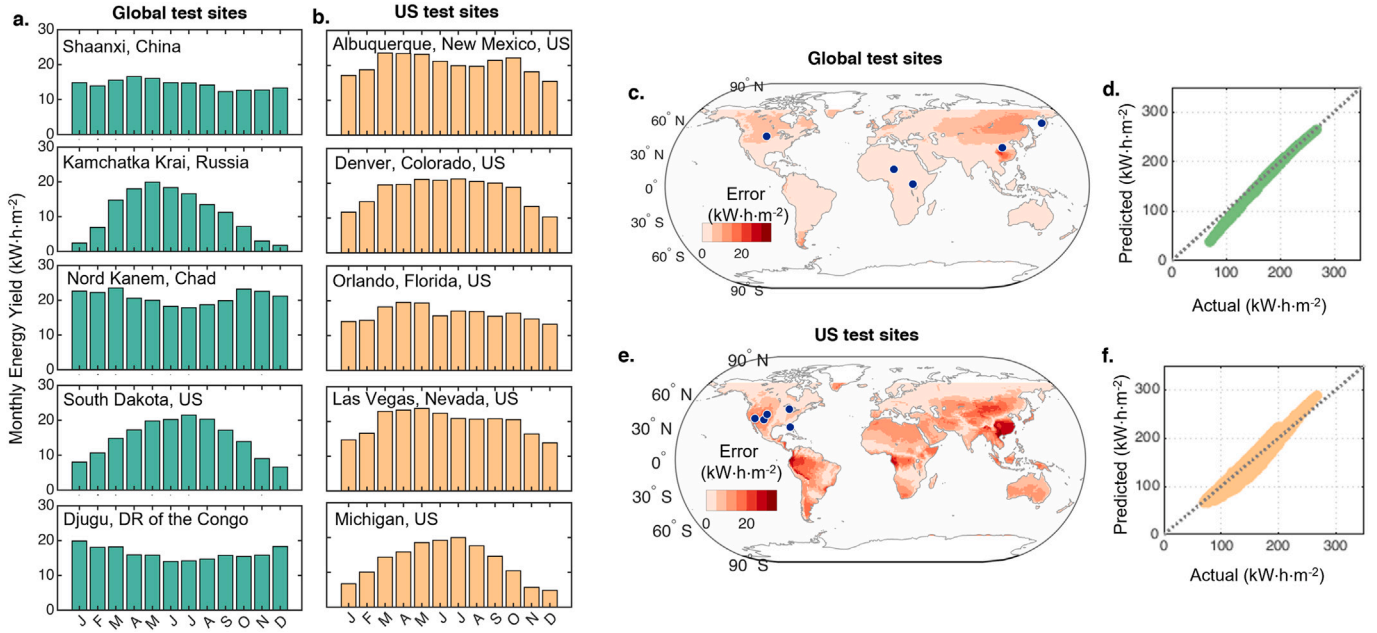


Fig. 3. Results with synthetic datasets. (a and b) Barplots showing the simulated monthly energy yields (M) used to construct the training data sets S_1 and S_2 . The prediction accuracy of models trained on data from (c and d) the global data set (S_1) and (e and f) DOE-NREL-Sandia's test site locations across the US (S_2). Maps in (c) and (e) show that, with monthly yield data from 5 locations (blue dots), the models can predict yearly EY at most locations with less than just 16 kWh m⁻² error. (d) and (f) shows that the models can explain most of the variability.

can effectively train an ML model to accurately predict Y potential across the world. These results are also consistent with our hypothesis that data from a small number of locations can sufficiently predict the global yield potential.

4. Heterogeneous crowd-sourced PV data: unsupervised experiments

Having demonstrated that a small, representative, high-quality synthetic dataset can predict the global YY potential, we now turn to utilizing already existing large, potentially low-quality outdoor PV performance data. We collected monthly energy yields of over 3000 PV systems from various public PV performance data repositories such as PVOutput and DuraMAT, online dashboards, and reports in the literature. The locations of the systems are shown in Fig. 2. Compared to the synthetic datasets, these data represent results from weakly controlled experiments. While such data may be available in large quantities, it is inherently heterogeneous. Although all systems employed monofacial modules, the systems varied in module attributes, design, and capacity. The nominal system capacities varied from a few kilowatts to megawatts (see SI Fig. S3). For many outdoor systems, parameters such as system capacity and tilt were unknown. The duration of the available data varied from just one year to multiple years.

4.1. Correcting for data heterogeneity

Given the heterogeneity, it is essential to verify whether the input variables can still adequately explain the output. For the heterogeneous systems, Eq. (1) can be reformulated as

$$M_{\text{field}} = f(I_{\text{GHI}}, T_{\text{amb}}, k_t, U) \quad (2)$$

where U is a set of unknown variables that also influence the yield, e.g., system capacity, system orientation.

The primary source of variability in Y_{field} is the variation in system capacities that leads to yield variations spanning four orders of magnitude. This disparity in system capacities can be corrected by scaling the measured Y_{field} at a location to equate that of a reference PV system with

known parameters at the same location. In the absence of existing physical reference systems, we simulated an optimally oriented system and use it as the reference (Y_{sim}). We assume that the Y_{field} and the Y_{sim} at a location is related by a scaling factor,

$$s = \frac{Y_{\text{field}}}{Y_{\text{sim}}} \quad (3)$$

s is a constant coefficient for a given system that accounts for the difference in system capacity. After scaling, the monthly yields of the outdoor system become

$$M_{\text{field}}^* = s \times M_{\text{field}} \quad (4)$$

M_{field}^* can be seen as the output of the reference system perturbed by some noise, i.e.,

$$M_{\text{field}}^* = M_{\text{sim}} + \epsilon \quad (5)$$

The noise term (ϵ) represents the discrepancy between M_{sim} and M_{field}^* owing to other unknown variables and model/measurement uncertainties. While s corrects for the differences in the system capacities in the dataset, further reducing ϵ would require information about additional variables to explain the variability. Although algorithms exist to also correct for the tilt and azimuth parameters [25], it is reasonable to assume that the field systems were professionally installed and had near-optimal orientation, i.e., tilt and azimuth, to maximize yield. Also, since some noise is inevitable in any outdoor measurement, such data represents a more practical scenario with weaker control over variability compared to the synthetic datasets discussed before.

After correcting for system capacity and computing the M_{field}^* for all field locations, additional filtering was performed to remove the anomalous sites (see SI). Finally, the training dataset F ($N = 36,000$) was created to train the PGML model.

4.2. Predicting the worldwide performance of solar farms

Fig. 4(a) shows the performance of the PGML trained on the entire homogenized dataset F . With the large number of sites in this dataset,

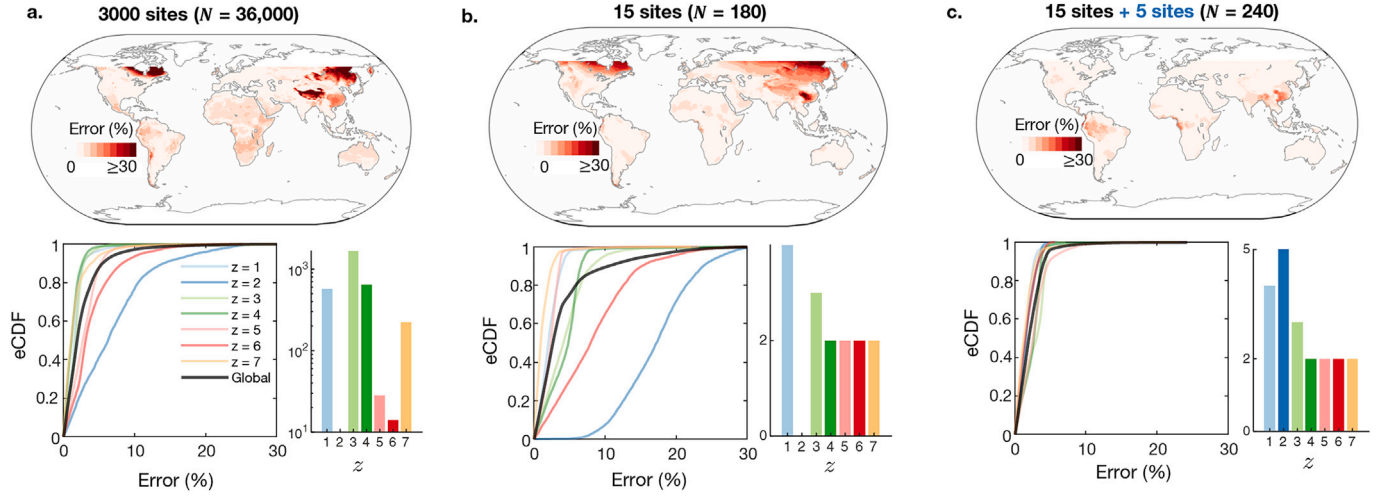


Fig. 4. Performance of the PGML model trained on homogenized public PV performance data, corresponding zone-wise (z) cumulative distributions (eCDF) of error. The histograms show the number of sites in each zone for each training dataset. (a) When trained on the entire homogenized dataset F , the error is less than 10 % for 95 % of the locations worldwide. Higher errors are observed at zones 2 and 6 due to their lack of representation in the dataset. (b) By choosing just 15 sites from F ensuring spatial diversity, similar accuracy can be achieved for all zones, except for zones 2 and 6. (c) By introducing synthetic data for 5 sites in zone 2, the error in these zones is significantly reduced resulting in less than 5 % error for 95 % of global locations.

we expect this model to perform well. The model demonstrates high accuracy consistent with this expectation: the prediction error was less than 10 % compared to a PGML trained on global simulations for 95 % of the global sites. However, we observe that the errors are higher at locations in zones 2 and 6 situated at higher latitudes. Such higher errors were expected since these two zones had the lowest number of observations in the training dataset, which is expected since zone 2 comprises regions near the Arctic Circle and the Himalayas that are largely uninhabited. These higher latitude regions experience extreme fluctuations in temperature over the months, making them unique. This shows that even with a large training dataset, the model may fail to generalize globally due to a lack of representation for certain regions.

Next, we test our hypothesis that a smaller subset of this large dataset, if *diversely sampled*, could predict the Y with a similar accuracy. For this, we chose 15 sites from F ensuring that each zone was represented, as shown in Fig. 4(b). We denote this smaller dataset by F_s ($N = 180$). Since no sites in F belonged to zone 2, it could not be represented. As the eCDF in Fig. 4(b) shows, despite having merely 0.5 % of the observations of F , the model achieves comparable accuracy (10 % error 95th percentile) for zones other than 2 and 6. However, the overall accuracy of the global model is limited by the least represented zones, i.e., zones 2 and 6, as higher errors in these zones lead to an overall 15 % error 95th percentile. Nevertheless, this result supports our hypothesis that a smaller dataset, when diversely sampled, contains sufficient information to make global-scale yearly yield prediction.

We can enhance this model by addressing the gaps in representation for specific zones, which clearly affect the accuracy of global predictions. By combining simulated results from 5 randomly chosen sites from zone 2 with F_s , we constructed a fused dataset F_s^* ($N = 240$). As shown in Fig. 4(c), when trained on F_s^* , the PGML outperformed F achieving an overall error of 6 % (95th percentile). These results further support the hypothesis that data from sparse geographical locations can train a model to predict worldwide annual yields of solar farms.

4.3. Validating against unseen PV field data

Now, we test the model's ability to predict real-world PV yield at various locations across the world. To demonstrate the performance of PGML when trained on F_s^* , we predict the M_{field}^* of several real-world systems across the world. The data from these systems were not directly included

Table 1

Comparison of the PGML predicted yields with simulated and field data.

Sites	PVZone	MAPE M_{field}	Y_{field}	MAPE M_{sim}
Ohio, USA	3	10.8 %	0.9 %	12.8 %
Cochin, India	7	3.8 %	0.3 %	4.4 %

in the training process. However, since the zones were represented in the training dataset, we expect the model to be able to generalize to these locations as well and predict the yearly yield. Fig. 5 shows the results for two sites located in Ohio, USA ($z = 3$) [26] and Cochin, India ($z = 7$) [27] as examples. Indeed, for both locations, the model can predict the field performance with good accuracy as shown in Fig. 5 and summarized in Table 1. M_{PGML} denotes the yield predicted by the PGML model trained on dataset F mentioned above. M_{field}^* and M_{sim} denote the scaled field data and PVMAPS simulated yield values, respectively. Ideally, we want these quantities to be as close to each other as possible. For the site in Ohio, the error for Y compared to Y_{field}^* was 1.21 kWh m⁻² (0.9 %), and the RMSE for M_{field}^* predictions was 3.44 kWh m⁻² (10.8 % MAPE). For the site in Cochin, the error for Y was 0.64 kWh m⁻² (0.3 %), and the RMSE for M_{field}^* predictions was 0.66 kWh m⁻² (3.8 % MAPE). Supplementary Fig. S6 compares the monthly yield profiles predicted by the PGML models for 14 additional sites not included in the training dataset. Images of some of these systems are shown in Supplementary Fig. S7, demonstrating the heterogeneity. These examples showcase the PGML model's ability to generalize to arbitrary locations if their zones are represented in the homogenized training data.

Looking at the examples in Fig. 5 and Fig. S6, we notice that field-collected homogenized monthly yields M_{field}^* do not always agree with M_{PGML} and even M_{sim} at every location. However, M_{PGML} and M_{sim} agree in nearly all locations during most months. Several factors contribute to the discrepancy (ϵ) beyond the inherent error associated with machine learning. We have assumed that the system parameters (e.g., capacity, orientation) of the systems remained constant throughout the data collection period. In reality, the active capacity could vary over months due to maintenance and repair, or simply faulty instruments, which would result in a significant change in the monthly yield profiles. Another source of error is the mismatch between actual inputs and

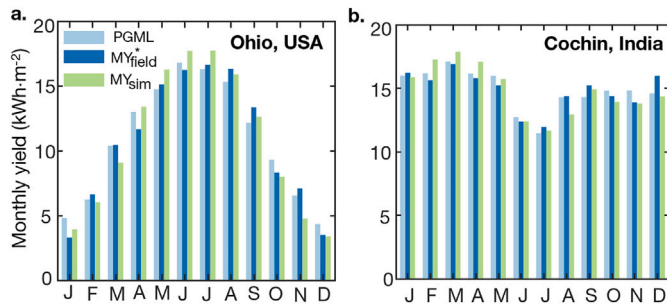


Fig. 5. Examples of M and Y predictions with the PGML model trained on public heterogeneous data for farms at (a) Ohio, USA and (b) Cochin, India. The PGML predicts the Y at these locations with just 0.9 % and 2 % relative errors compared to physics-based simulations (Y_{sim}), demonstrating the efficacy of the method when trained on a homogenized, representative dataset.

the climatic average values assumed during training. Since the data was available for a year for many of these locations, the input values could vary from the climatic average. However, such variabilities in the system parameters and inputs can be accounted for in coordinated controlled experiments. Nevertheless, the strong agreement between M_{PGML} and M_{sim} indicates that the PGML model can be robust against these variabilities with large enough observations from each zone. Since public field data is naturally susceptible to these variabilities, increasing the number of observations may help improve the accuracy of monthly yield predictions.

5. Discussion

The above results underscore the critical importance of ensuring data diversity in training the PGML models. Since the PGML model essentially solves a regression problem, the generalization power of the model strongly relies on how well its training dataset covers the input domain. The paper demonstrates that PVZones serves as a powerful tool for generating representative datasets by systematically accounting for gaps in representation. Despite its size, the dataset built from field-performance data, F , lacked representation for zone 2 locations experiencing low monthly average temperatures ($< -5^\circ\text{C}$), which explained the observed lower accuracy for zone 2 in Fig. 4(a). The gaps in field observations can be addressed through data fusion. The fused dataset, F_s^* , which includes a small amount of simulated data for zone 2, significantly improves the model's global performance. The gaps in representativeness can also be easily visualized by the parallel plots shown in Fig. 1(c–d).

Recently, there has been growing interest in data-centric AI models that enhance ML performance while reducing data requirements to achieve strong generalization. This work demonstrates that in the context of PV, diverse sampling from physically relevant variables reduces the need for large datasets—which are costly to obtain across numerous locations—while maintaining a greater degree of experimental control. By doing so, one can also reduce the number of parameters required to learn the solution space, resulting in smaller, data-efficient models. The PGML model for predicting solar farm yield can be trained and used to infer yield globally within seconds on a modern personal computer. Due to their speed, such fast statistical models can complement more comprehensive physics-based models to check for physical consistency, as demonstrated by their use in homogenization. This enables us to create computationally efficient and accessible tools to guide policy decisions for regional collaborations, determine the best sites for creating national/regional test centers, and cross-check predictions provided by commercial and costly detailed analysis based on proprietary software. The generated datasets and the developed models can be downloaded for free [24].

To summarize, we have presented a method to assimilate global patterns in the energy yield potential of solar farms with data from

strategically chosen locations. Achieving this requires training datasets that accurately represent the domain of input variables. Toward this end, we have presented the concept of problem-specific climate maps to facilitate the construction of representative datasets. The trained models utilizing simulated and actual field-measured data accurately ($\leq 6\%$ error) predicted the yield potentials obtained through detailed worldwide physics-based simulations. The results lead to the following key conclusions.

1. Physical understanding should guide the choice of relevant variables for a PGML model. When predicting the energy yield potential of a PV system, the key determining variables are I_{GHI} , T_{amb} , and k_t . Additional variables can be introduced based on their physical significance in determining the target variable, such as surface albedo for bifacial systems.
2. A large quantity of data alone is insufficient to predict the Y of PV systems worldwide. For accurate prediction across the globe, the dataset needs to be representative. We have shown that PV-specific climate maps (like PVZones) of physically relevant variables are a powerful tool to identify the different regions that need to be represented to construct a representative dataset. We have demonstrated that high-quality training data, obtained from a small number of strategically selected test sites, can predict global energy yield with very high accuracy.
3. Since the methodology can be extended to any PV technologies and arbitrary farm topologies, strategically placed test sites provide a powerful empirical means to gauge the potential of the novel technologies. Thus, our results define an innovative data-driven, physics-based machine learning model for creating resources to accelerate the worldwide adoption of PV technologies.

The PGML model trained on yield potential data can be used to monitor the health of the PV systems over time, as the predictions represent the expected performance. However, attribution (degradation or change in system parameters) will require further analysis. Furthermore, the approach is not limited to predicting solar farm performance; it can be adapted to a wide range of problems where the input space of physically significant variables is appropriately partitioned. We hope our approach will inspire broader applications of data-efficient machine learning models to accelerate the decision-making process for PV deployment.

CRedit authorship contribution statement

Jabir Bin Jahangir: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Muhammad Ashraf Alam:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Climate data. Climatic normal dataset containing the monthly averages of observations since 1984 was obtained from NASA POWER. The

global dataset has a spatial resolution of $0.5^\circ \times 0.5^\circ$ and consists of data for over 250,000 locations.

Constructing PVZones for energy yield. The zones in the PVZones map (shown in Fig. 1(b)) were determined by k-means clustering of physically relevant variables. To predict energy yields, the primary variables are I_{GHI} , T_{amb} , and k_t . The clustering was performed with the annual average values of these variables obtained from the POWER dataset.

Global physics-based solar farm simulation. To test and validate the machine learning model, we used the PVMAPS [24] to simulate the reference worldwide yearly yield potential of monofacial solar farms. Various aspects of the physical model are described in detail in Ref. [16]. We simulated systems employing monofacial modules ($\eta = 24\%$) installed in a fixed tilt configuration. During the simulation, the array pitch (p) was assumed to be 3 m and the module height (h) and elevation were assumed 1 m. The modules were optimally tilted to maximize the annual power output. The physical simulation is performed on a coarse grid with a spatial resolution of $2^\circ \times 18^\circ$ (1200 locations) on a monthly basis for a year with climatological normal data. Finally, the reference yearly yields (shown in Fig. 2) were calculated using the method described in SI Sec. S1.

Global model error. The error of the PGML model trained on a dataset is computed by comparing the predicted the Y on a regular $0.5^\circ \times 0.5^\circ$ grid using the following equation.

$$E = \frac{|Y_{\text{pred}} - Y_{\text{pred}}^*|}{Y_{\text{pred}}^*} \times 100 \quad (6)$$

Y_{pred} is the yield predicted by the PGML model, and Y_{pred}^* is the yearly yield predicted by a reference PGML trained on gridded simulation data. Y_{pred}^* is shown in SI Fig. S1.

Public PV performance data. Outdoor PV performance data for monofacial PV systems are collected from public dashboards and literature from over 3000 different PV systems around the globe (see Fig. S2). If available, M_{field} data from multiple years were averaged. Raw M_{field} data were filtered using the procedure described in SI Fig. S4. First, raw M_{field} data was scaled to with factor s to obtain M_{field}^* (in kWh m^{-2}) as described in the main paper. Then, RMSE was computed with respect to the monthly yields predicted by the physics-based simulation (M_{sim}) for the same location. Finally, sites with RMSE greater than the threshold e_{th} (marked red) were removed. e_{th} was set to 3 standard deviations of the overall RMSE distribution.

Appendix B. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.apenergy.2025.126192.

Data availability

Data will be made available upon request.

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